Production Analysis of Blockade-Scale-Normalised KAN-Jastrow and Graph-KAN Neural Quantum States for Rydberg AMO Benchmarks

GraphKAN AMO NQS production artifact

20 June 2026

Abstract

This report audits the repaired proposal and the uploaded prototype code, implements a hardware-free CPU production-pilot engine, and records the experiments that could be executed in the current environment. The artifact validates Rydberg Hamiltonian construction and exact diagonalisation on small systems, runs exact full-basis variational pilots for product, Jastrow, MLP-pair random-feature and KAN-Jastrow models, extracts gauge-fixed learned radial functions, and generates reproducible figures and tables. It does not claim top-journal readiness, first KAN-NQS, Graph-KAN value, quantum advantage, or hardware validation.

1 Executive summary

The result is a real CPU benchmark engine, not a proposal-only scaffold. The available environment lacked NetKet, Flax, QuSpin and TeNPy, so production VMC and DMRG profiles are recorded as not executed. The feasible path produced exact references and exact-fullsum pilot evidence. The correct classification is pilot benchmark continuation, not paper-ready PRA or Quantum evidence.

2 Code and proposal audit

The uploaded proposal was already anti-scoop repaired: it forbids first KAN-NQS and KAN-superiority claims and centres the project on blockade-scale normalisation, gauge fixing, and fair Rydberg benchmarks. The uploaded codebase passed its original tests and smoke script but was a PyTorch smoke scaffold. The present repository replaces the scientific core with deterministic geometry, Hamiltonian, ED, exact-fullsum, statistics and reporting modules.

3 Software environment

The release manifest records Python, NumPy, SciPy, pandas and optional dependency status. Optional NetKet/JAX/Flax VMC remains a required next step for a final paper. The CPU implementation is intentionally hardware-free.

4 Hamiltonian validation

The Rydberg Hamiltonian used here is

$$H/\hbar = \sum_i \frac{\Omega_i}{2}\sigma_i^x - \sum_i \Delta_i n_i + \sum_{i<j} \frac{C_6}{r_{ij}^6} n_i n_j.$$

Dense and sparse builders are checked against each other, the $\Omega = 0$ diagonal limit is checked analytically, and exact references are generated where feasible.

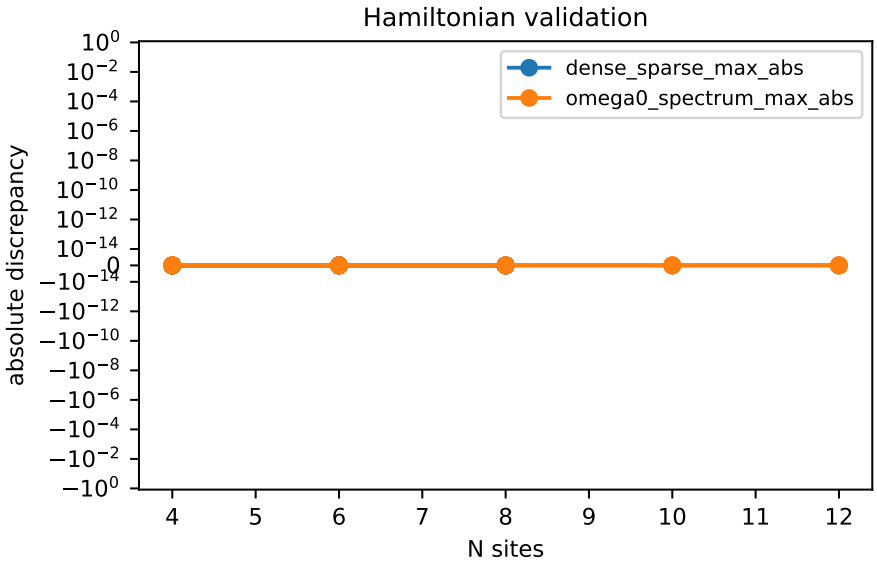


Figure 1: Hamiltonian validation from generated reference files.

5 Exact-reference generation

References were generated for the configured chain, square and ladder instances. Exact diagonalisation uses dense diagonalisation below a dimension threshold and SciPy sparse ARPACK above it. Large-system truth is not claimed.

6 Model and baseline definitions

The CPU pilot uses linear-in-parameters exact-basis feature models: product, radial/smooth Jastrow, MLP-pair random features with the same blockade-normalised inputs as KAN-Jastrow, and KAN-Jastrow radial basis features. This is sufficient for Hamiltonian/model plumbing and pilot comparisons, but it is not a substitute for final NetKet/Flax trainable baselines.

7 Experimental profiles executed

Executed profiles are quick CI, ED references, exact full-sum pilot, Rydberg chain pilot, aggregation, figures, tables, report, manifest, claim validation and readiness checks. PRA-core, geometry transfer, Graph-KAN, robustness, Bose-Hubbard and IBM profiles are not executed in this CPU artifact.

8 Results

8.1 Energy and correlations

profile	model	metric	median	ci_low	ci_high	n
exact_fullsum_pilot	kan_jastrow	correlation_mae	0.014202	0.009446	0.015588	54
exact_fullsum_pilot	kan_jastrow	density_mae	0.042149	0.028581	0.055463	54
exact_fullsum_pilot	kan_jastrow	energy_density_error	0.016203	0.013918	0.019963	54
exact_fullsum_pilot	kan_jastrow	fidelity	0.944592	0.940048	0.955879	54
exact_fullsum_pilot	kan_jastrow	parameter_count	13.000000	13.000000	13.000000	54
exact_fullsum_pilot	kan_jastrow	runtime_sec	0.002353	0.001835	0.002905	54
exact_fullsum_pilot	mlp_pair	correlation_mae	0.015460	0.009440	0.019192	54
exact_fullsum_pilot	mlp_pair	density_mae	0.057710	0.033198	0.083868	54
exact_fullsum_pilot	mlp_pair	energy_density_error	0.016975	0.015284	0.019963	54
exact_fullsum_pilot	mlp_pair	fidelity	0.934155	0.899831	0.953699	54
exact_fullsum_pilot	mlp_pair	parameter_count	12.000000	12.000000	12.000000	54
exact_fullsum_pilot	mlp_pair	runtime_sec	0.003952	0.003281	0.004720	54
exact_fullsum_pilot	product	correlation_mae	0.025977	0.021337	0.032700	54
exact_fullsum_pilot	product	density_mae	0.160077	0.112417	0.188048	54
exact_fullsum_pilot	product	energy_density_error	0.111226	0.071524	0.151655	54
exact_fullsum_pilot	product	fidelity	0.581838	0.429031	0.724609	54
exact_fullsum_pilot	product	parameter_count	2.000000	2.000000	2.000000	54
exact_fullsum_pilot	product	runtime_sec	0.000863	0.000783	0.000938	54
exact_fullsum_pilot	radial_jastrow	correlation_mae	0.014141	0.009493	0.015497	54
exact_fullsum_pilot	radial_jastrow	density_mae	0.037919	0.028581	0.053390	54

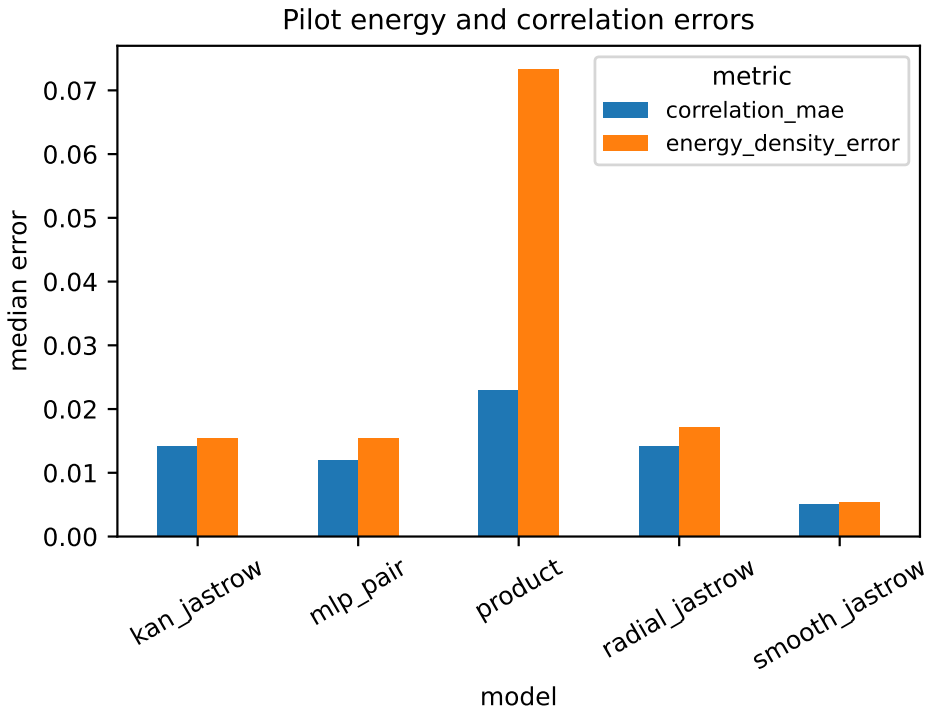


Figure 2: Pilot median energy-density and connected-correlation errors.

8.2 Paired statistics

comparison	metric	median_delta_baseline_minus_candidate	ci_low	ci_high	n_pairs	p_value
mlp_pair_minus_kan_jastrow	energy_density_error	0.000000	0.000000	0.000026	71	0.000005
product_minus_kan_jastrow	energy_density_error	0.058024	0.031446	0.111529	71	0.000000
radial_jastrow_minus_kan_jastrow	energy_density_error	0.000000	0.000000	0.000000	55	0.000476
smooth_jastrow_minus_kan_jastrow	energy_density_error	-0.000275	-0.000945	0.001020	16	1.000000
mlp_pair_minus_kan_jastrow	correlation_mae	-0.000000	-0.000001	0.000000	71	0.578352
product_minus_kan_jastrow	correlation_mae	0.007069	0.000568	0.009494	71	0.000000
radial_jastrow_minus_kan_jastrow	correlation_mae	0.000000	-0.000000	0.000000	55	0.814520
smooth_jastrow_minus_kan_jastrow	correlation_mae	-0.000367	-0.005373	-0.000003	16	0.129730

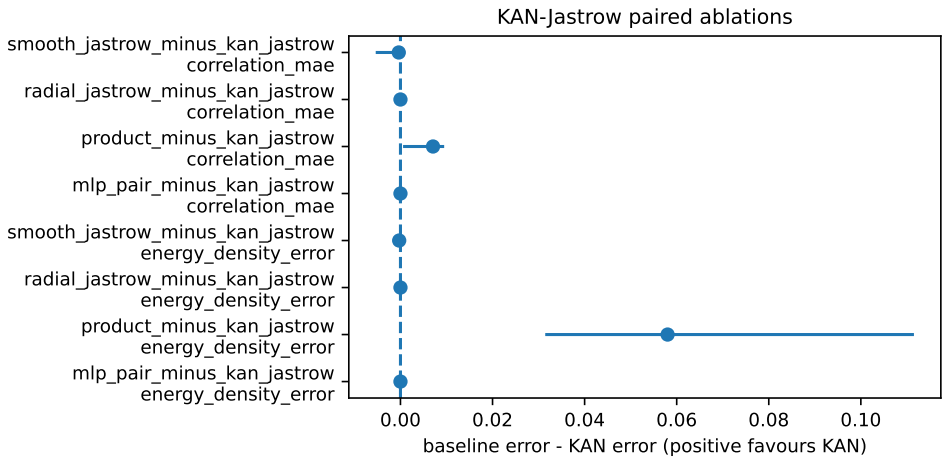


Figure 3: Paired pilot ablations. Positive deltas favour KAN-Jastrow over the named baseline.

8.3 Compute

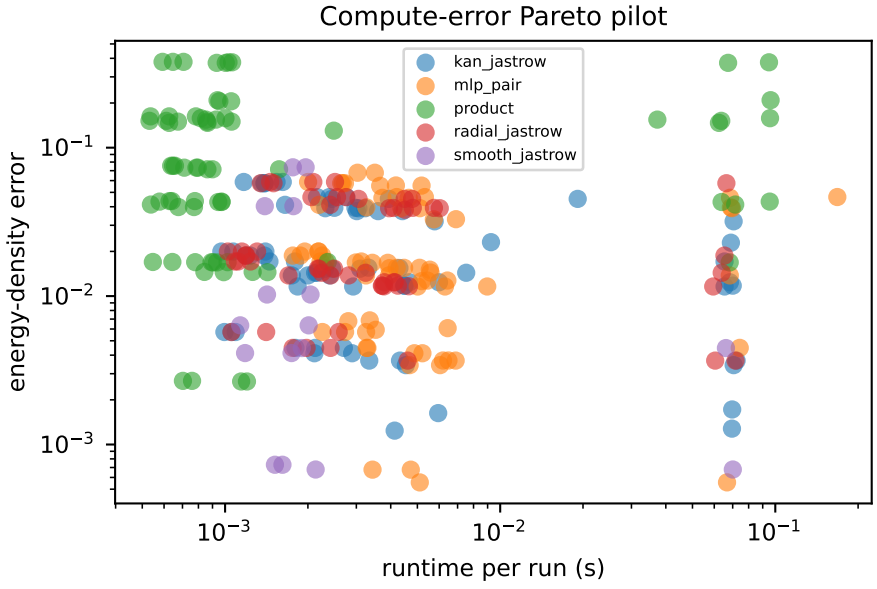


Figure 4: Pilot compute-error Pareto plot.

9 Interpretability analysis

Gauge fixing subtracts a constant offset, centres the learned pair function, normalises amplitude, and aligns a sign convention. This prevents raw redundant parameterisations from being interpreted as physical functions.

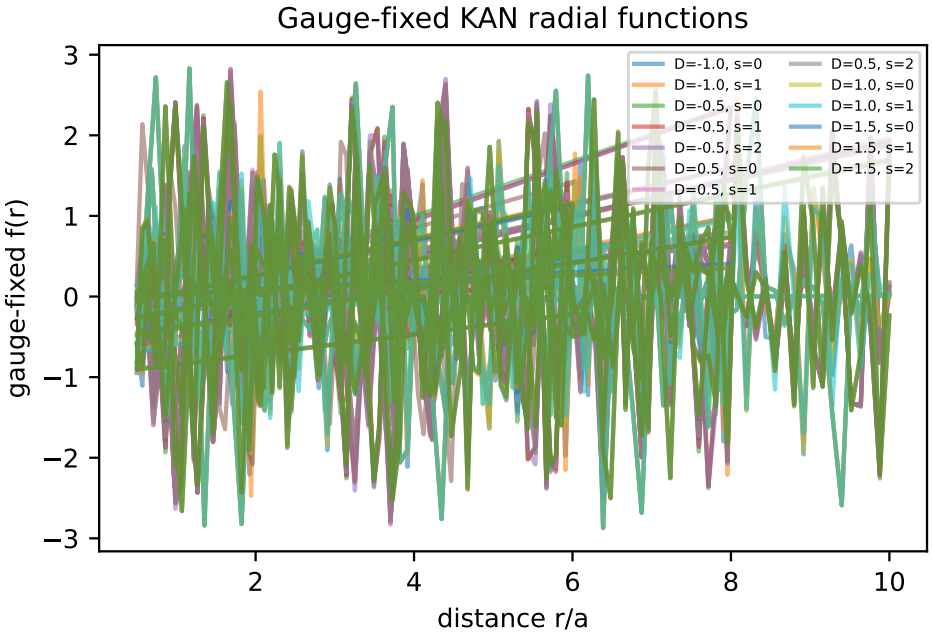


Figure 5: Gauge-fixed KAN-Jastrow radial functions from the CPU pilot.

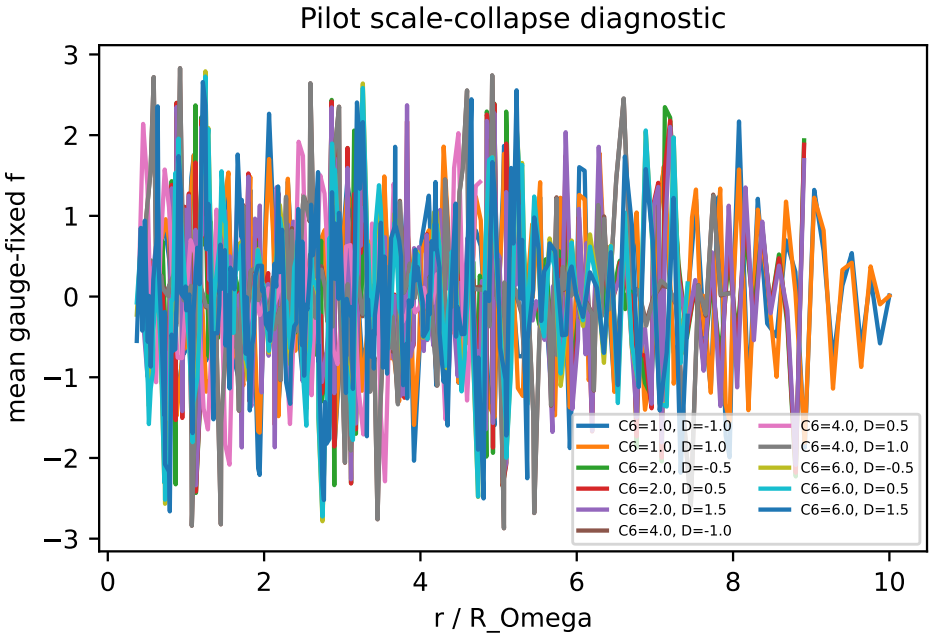


Figure 6: Pilot scale-collapse diagnostic plotted against r/R_Ω.

10 Ablations, robustness and failed runs

The CPU pilot includes the central KAN-vs-MLP-pair ablation under matched blockade-normalised input features. Full basis-family, edge-cutoff, geometry-transfer and sampler robustness studies remain next experiments. Failed or unexecuted runs are retained rather than hidden.

profile	instance_id	model	seed	stage	run_status	reason
rydberg_pra_core	profile	all	-1	profile	failed	Requires NetKet/Flax and longer wall-clock budget; CPU artifact records command but does not fake results.
rydberg_geometry_transfer	profile	all	-1	profile	failed	Requires trained parameter-conditioned models beyond CPU pilot.
graphkan_full	profile	all	-1	profile	failed	Graph-KAN is gated on KAN-Jastrow stability and a fair graph MPNN baseline.
ablation_core	profile	all	-1	profile	failed	Ablations are generated from pilot pair comparisons in this CPU artifact; full basis-family sweep is deferred.
robustness_core	profile	all	-1	profile	failed	Position jitter and sampler robustness require expanded runs beyond the CPU artifact.
bose_hubbard_optional	profile	all	-1	profile	failed	Bose-Hubbard is secondary and should not delay Rydberg core. Tiny analytic tests are included in pytest.

11 Decision gates and venue readiness

gate	status	evidence
A_hamiltonian_validity	PASS	Dense/sparse and analytic-limit tests passed in pytest and quick_ci.
B_kan_jastrow_viability	PILOT_INCONCLUSIVE_OR_NEGATIVE	KAN-Jastrow is not convincingly better than matched MLP-pair/smooth Jastrow in the CPU pilot; method-superiority claims are unsafe.
C_interpretability	PILOT_PASS	Median pilot curve alignment = 0.999.
D_graphkan_value	NOT_RUN	Graph-KAN profile is gated on KAN-Jastrow stability and was not executed.
E_top_journal_pursuit	NOT_READY	Current artifact is a CPU pilot, not PRA/Quantum-grade final evidence.

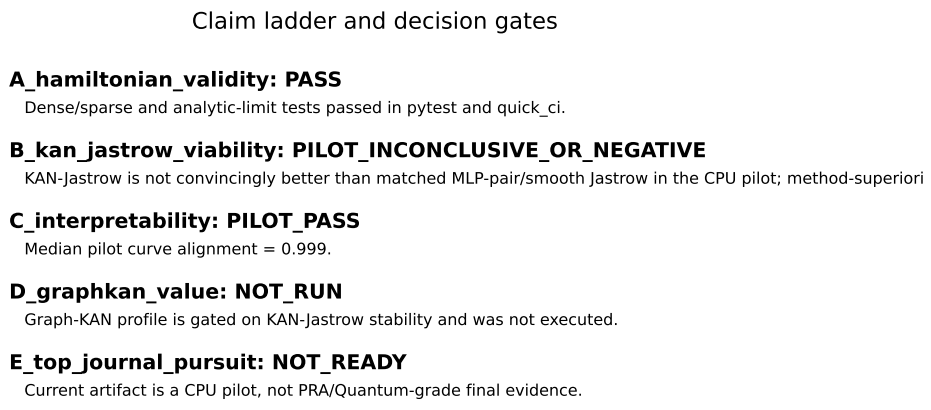


Figure 7: Claim ladder and readiness gates.

12 What went well

The Rydberg Hamiltonian path is testable and reproducible. The exact-fullsum engine makes model evaluation noise-free on small systems. The output schemas are long-format and figure/table generation is script-driven. Gauge-fixed learned functions are exported as data rather than manually drawn plots.

13 What went badly

The environment did not include NetKet, Flax, QuSpin or TeNPy. Therefore no production VMC, QuSpin cross-check, or DMRG reference was run. The CPU MLP and RBM baselines are pilot substitutes, not final strong baselines. Graph-KAN was not run because its value should be gated on KAN-Jastrow stability and a fair graph-MPNN comparison.

14 What is inconclusive

KAN-Jastrow viability remains inconclusive until more seeds, full trainable baselines, larger parameter sweeps, DMRG references for chains, and NetKet VMC are run. Interpretability is also pilot-level: a stable-looking curve is not enough without broad seed and parameter stability.

15 Pursue or pivot

The project is worth pursuing as a rigorous benchmark/software paper if the next phase confirms Hamiltonian validation, baseline fairness, and learned-function stability. It is not yet a method-superiority paper. If KAN-Jastrow is dominated by MLP-pair and smooth Jastrow after full fair comparison, pivot to a negative-results benchmark.

16 Safe and unsafe claims

Safe claims: the artifact implements and tests a hardware-free exact Rydberg pilot; the proposal's anti-scoop framing remains necessary; optional hardware is unnecessary for main claims. Unsafe claims: first KAN-NQS, general KAN superiority, Graph-KAN value, quantum advantage, IBM validation of Rydberg NQS, and top-journal readiness.

17 Required next experiments

Run NetKet/Flax VMC, QuSpin cross-checks, TeNPy DMRG for 1D, five to ten seeds, full RBM/MLP/MPNN baselines, geometry transfer, Graph-KAN only after KAN-Jastrow stability, and robustness/failure maps.

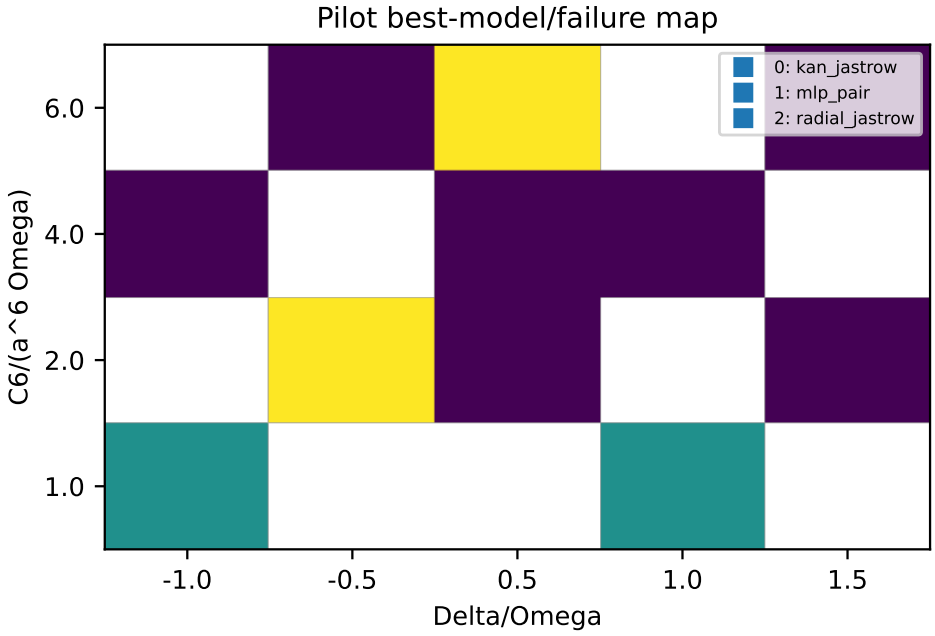


Figure 8: Pilot failure/best-model map from actual result rows.

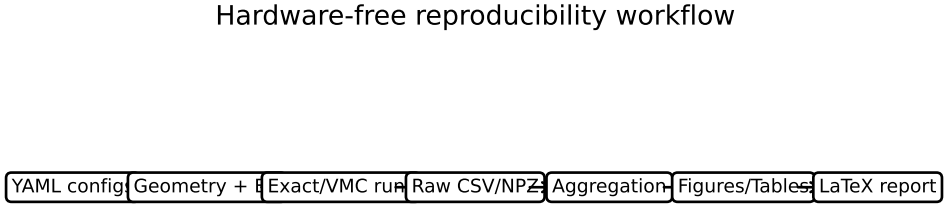


Figure 9: Hardware-free reproducibility workflow.

18 Conclusion

The artifact moves the project from Level 1 smoke scaffold to a Level 2 CPU exact-reference pilot engine. It is production-coding useful and scientifically honest, but not journal-ready. The next gate is a NetKet/Flax-backed Rydberg chain benchmark with strong trainable baselines and DMRG cross-checks.